Install Guide for Intake Scanning App

Pre-requisites

Ensure the following software and hardware are available:

- **Mac Computer** with macOS 12.0 or later (required for iOS development)
- **Xcode 15+** (Download Xcode (https://developer.apple.com/xcode/))
- **Python 3.9+** (Download Python (https://www.python.org/downloads/))
- **Firebase Project** set up (Firebase Console (https://console.firebase.google.com/))
- **Apple Developer Program Membership** (Join here (https://developer.apple.com/programs/))

Firebase Setup

1. **Create a Firebase Project:**
   - Go to the Firebase Console (https://console.firebase.google.com/).
   - Click **Add Project** and follow the prompts.
2. **Enable Authentication:**
   - In the Firebase Console, go to **Authentication**.
   - Enable **Email/Password** sign-in method.
3. **Create a Realtime Database:**
   - Go to **Build > Realtime Database**.
   - Create a database and start in test mode.
4. **Enable Storage:**
   - Go to **Build > Storage**.
   - Create a default storage bucket.
5. **Generate Admin SDK Key:**
   - Go to **Project Settings > Service Accounts**.
   - Click **Generate new private key** and download the JSON file.
   - Place this file (`jib-4338-hisa-firebase-adminsdk-c9uu0-6b2a83c10a.json`) into your backend directory.

6. **Configure Firebase Rules:**
    - Update Realtime Database and Storage rules to allow read/write during development.

Example Development Rules:

```
{
  "rules": {
    ".read": true,
    ".write": true
  }
}
```

> ⚠ *Note: Update rules to more secure settings if making a production build!*

Dependent Libraries

The project uses the following main dependencies:

Backend (Python):

```
pip install -r requirements.txt
```

Main packages:

- Flask
- firebase-admin
- torch
- opencv-python
- albumentations
- ultralytics (YOLOv8)

iOS App (Swift, via Swift Package Manager):

- FirebaseAuth
- FirebaseFirestore
- FirebaseStorage

Add Firebase to your iOS project through Swift Package Manager:

- URL: `https://github.com/firebase/firebase-ios-sdk`

Download Instructions

- Clone this repository:

```
git clone https://github.com/LukGH4/JIB-4338-HISA.git
```

Build Instructions

Backend

1. Navigate to backend directory:

```
cd backend
```

2. Create a virtual environment (optional but recommended):

```
python3 -m venv venv
source venv/bin/activate
```

3. Install dependencies:

```
pip install -r requirements.txt
```

4. Place your **Firebase Admin SDK key** (`jib-4338-hisa-firebase-adminsdk-c9uu0-6b2a83c10a.json`) into the backend root folder.

5. Place your YOLOv8 model file (`yolov8n_trained.pt`) into the backend root folder.

> ⚠ ***Important:*** *Update the `send_email()` function in `app.py` with a real Gmail address and App Password for sending email alerts. You must enable [App Passwords in Gmail (https://support.google.com/accounts/answer/185833?hl=en)](https://support.google.com/accounts/answer/185833?hl=en) if using Gmail.*

iOS App Setup (Xcode)

1. Open the `.xcodeproj` or `.xcworkspace` file.
2. Install Firebase SDK via Swift Package Manager:
   - Open **Xcode > File > Add Packages...**
   - Enter: `https://github.com/firebase/firebase-ios-sdk`

    - Add **FirebaseAuth**, **FirebaseFirestore**, and **FirebaseStorage**.
3. Add the downloaded **GoogleService-Info.plist** to your project root.
4. Update the **Bundle Identifier** in Xcode to match your Firebase iOS app settings.
5. Set **iOS Deployment Target** to 15.0 or higher.
6. Add usage descriptions to `Info.plist`:

```
<key>NSPhotoLibraryUsageDescription</key>
<string>Access to your photo library is needed for classifying media.</string>
<key>NSCameraUsageDescription</key>
<string>Access to your camera is needed for uploading photos and videos.</string>
```

7. Ensure your app targets real devices (for camera/photo access).

Installation of Application

- Create a folder called `/uploads` inside the backend directory (already auto-created if missing).
- Backend server listens on **port 3333** (`http://localhost:3333`).

Run Instructions

Backend

Run the Flask server locally:

```
python app.py
```

Server should start at `http://0.0.0.0:3333/`.

iOS App

- Select a device and **Run** in Xcode.
- Login/signup with Firebase Authentication.
- Begin uploading photos or videos.

Troubleshooting

Issue	Solution
YOLO model not found	Ensure `yolov8n_trained.pt` is in backend root
Firebase connection errors	Confirm `serviceAccountKey.json` is correct and database/storage rules are permissive

Issue	Solution
App build fails	Check Firebase SDK is properly linked, and make sure to check if the bundle ID is correct
iOS app cannot access Photos or Camera	Ensure proper permissions in `Info.plist`

Publishing the iOS App (If needed)

1. Set Up App Store Connect

- Create a new app in [App Store Connect (https://appstoreconnect.apple.com/)](https://appstoreconnect.apple.com/).
- Set the Bundle ID to match Xcode settings.
- Fill metadata (name, description, keywords, screenshots).

2. Prepare App for Release

- Set `iOS Deployment Target` (e.g., iOS 15.0 or higher).
- Use **Product > Archive** to create a build.
- Validate and Upload to App Store Connect.

4. Submit for Review

- Complete App Store Review submission.
- Respond to any issues flagged by Apple.
- Release upon approval!

> ⚡ ***Note:*** *Apps using Firebase/Auth must have a proper Privacy Policy linked in App Store metadata.*